

Fact Sheet: The Global Burden of Traumatic Brain Injury

The Scale of the Problem

An estimated 20.8 million new traumatic brain injuries occur each year worldwide, with 37.9 million people living with TBI at any given time. TBI is a leading global cause of death and disability across all age groups.

While age-standardized incidence rates have declined over the past three decades, the absolute number of TBI cases continues to rise, driven by population growth, road traffic expansion, and urbanization in low- and middle-income countries (LMICs).

Who Is Affected

TBI disproportionately impacts young males in LMICs, where road traffic accidents are the leading cause. Falls are the leading cause globally, but road injuries dominate in South Asia, Sub-Saharan Africa, and Southeast Asia — regions with the fastest-growing populations and the least-developed trauma systems.

An estimated 69 million people sustain TBI each year, with the highest prevalence in LMICs. Low-SDI (Sociodemographic Index) regions are experiencing rising prevalence and years lived with disability even as incidence rates fall, meaning more people are surviving TBI but living with its consequences.

Mortality

TBI-related mortality in LMICs is 3- to 4-fold higher than in high-income countries. A meta-epidemiological study of over 59,000 patients across 31 LMICs found a pooled TBI-related mortality of 16.7%. Within LMICs, mortality is significantly associated with poverty-related factors: median income, percentage of the population below the poverty line, and primary school enrollment rates.

The Invisible Burden

TBI is increasingly recognized as a chronic condition, not a single event. Survivors face long-term cognitive impairment, depression, post-traumatic epilepsy, and reduced quality of life. In a Ugandan cohort, more than 15% of TBI survivors continued to suffer from seizures and depression years after injury. Post-discharge mortality was 9% even among those who survived hospitalization.

The chronic burden of TBI is largely invisible in LMICs, where outcome data are not reliably recorded and long-term follow-up is rare. The true disability burden is almost certainly underestimated.

Leading Causes of TBI Worldwide (2021)

- Falls
- Road traffic injuries
- Interpersonal violence
- Exposure to mechanical forces
- Conflict and terrorism (the most rapidly increasing risk factor)

Key Takeaway

TBI is not just an acute emergency - it is a lifelong condition with cascading medical, social, and economic consequences. Most of this burden falls on populations with the least access to prevention, acute care, and rehabilitation. Every phase of the TBI care continuum - from prevention through long-term support - represents an opportunity for innovation.